

SIMATS ENGINEERING

ASSESSMENT TOOL 2

**COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY COURSE
CODE: CSA51**

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

Total Marks: 50

Annexure A

Code of Conduct:

I _____Varsha.S/192511358_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

1. Secure Password Storage in Web Applications

A web application stores user passwords and needs to protect them against database breaches.

- A. Explain why storing plaintext passwords is insecure.
- B. Suggest a secure method for storing passwords and justify your choice.

2. Digital Signature for Document Authentication

A company wants to ensure that official documents sent electronically are authentic and unaltered.

- A. Explain how digital signatures can be used in this scenario.
- B. Justify how this method ensures both integrity and non-repudiation.

3. Preventing Replay Attacks in Online Transactions

An online payment system is vulnerable to replay attacks where attackers resend previously captured transactions.

- A. Explain how replay attacks occur.
- B. Suggest a cryptographic technique to prevent such attacks and justify your answer.

4. Secure Communication over Public Wi-Fi

A user accesses sensitive information using a public Wi-Fi network without encryption.

- A. Identify the risks involved in this scenario.
- B. Recommend a secure communication method and justify how it protects user data.

5. Data Integrity in Cloud Storage

An organization stores critical data in a cloud server and wants to ensure that the data is not altered.

- A. Explain how hashing can be used to verify data integrity.
- B. Suggest an additional mechanism to enhance trust and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
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Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

- **Question 1: Secure Password Storage in Web Applications**
- **A. Why is storing plaintext passwords insecure?**

Storing passwords in plaintext means the passwords are saved exactly as entered by users. If the database is hacked, attackers can immediately view and misuse all passwords.

Risks of plaintext password storage:

- Passwords are exposed during database breaches.
- Attackers can log in to user accounts without any effort.
- Users often reuse passwords on multiple websites, leading to credential theft.
- Organizations lose customer trust and may face legal consequences.

B. Suggest a secure method for storing passwords and justify your choice.

The most secure method is to store **hashed passwords with a unique salt** using algorithms such as **bcrypt, Argon2, or PBKDF2**.

Working

1. User enters a password.
2. A random salt is generated.
3. The password and salt are combined.
4. The combined value is hashed.
5. Only the hash is stored in the database.

During login, the entered password is hashed again with the same salt and compared with the stored hash.

Advantages

- Original passwords cannot be recovered from the hash.
- Salt prevents rainbow table attacks.
- Strong hashing algorithms resist brute-force attacks.
- Improves overall system security.

Conclusion

Using salted password hashing protects user credentials even if the database is compromised.

• **Question 2: Digital Signature for Document Authentication**

A. Explain how digital signatures can be used.

A digital signature verifies that an electronic document is authentic and has not been modified.

Working

1. The sender calculates the hash of the document.
2. The hash is encrypted using the sender's private key.
3. The encrypted hash becomes the digital signature.
4. The document and signature are sent to the receiver.
5. The receiver decrypts the signature using the sender's public key.
6. The receiver computes the document hash again.
7. If both hashes match, the document is authentic.

B. How does it ensure integrity and non-repudiation?

Integrity

- Any change in the document changes its hash.
- If the hashes differ, the document has been modified.

Non-repudiation

- Only the sender owns the private key.
- Therefore, the sender cannot deny signing the document.

Advantages

- Verifies document authenticity.
- Detects unauthorized modifications.
- Provides legal proof of authorship.

Conclusion

Digital signatures ensure authentication, integrity, and non-repudiation for electronic documents.

- **Question 3: Preventing Replay Attacks in Online Transactions**
- **A. Explain how replay attacks occur.**

A replay attack happens when an attacker captures a valid transaction message and resends it later to perform unauthorized actions.

Example

A customer transfers ₹500 online.

An attacker captures the encrypted transaction and retransmits it.

The bank processes the same transaction again, resulting in duplicate payments.

Risks

- Financial loss.
- Unauthorized transactions.
- Loss of user trust.
- Data misuse.

B. Suggest a cryptographic technique to prevent replay attacks.

The best solution is to use **nonces, timestamps, and message authentication codes (MACs)**.

Working

- Every transaction includes a unique nonce or timestamp.
- The server checks whether the nonce has already been used.
- Duplicate requests are rejected.
- A MAC verifies that the message has not been modified.

Advantages

- Prevents duplicate transactions.
- Ensures message freshness.
- Protects transaction integrity.
- Improves authentication.

Conclusion

Using nonces, timestamps, and MACs effectively prevents replay attacks in online payment systems.

• Question 4: Secure Communication over Public Wi-Fi

A. Identify the risks involved.

Public Wi-Fi networks are generally insecure because data can be intercepted by attackers.

Risks

- Eavesdropping.

- Man-in-the-Middle (MITM) attacks.
- Password theft.
- Session hijacking.
- Data modification.
- Identity theft.

B. Recommend a secure communication method and justify your answer.

The recommended solution is to use **HTTPS (TLS/SSL)** or a **Virtual Private Network (VPN)**.

HTTPS

- Encrypts communication between the browser and server.
- Prevents attackers from reading transmitted information.
- Verifies server authenticity using digital certificates.

VPN

- Creates an encrypted tunnel between the user's device and the VPN server.
- Protects all internet traffic on public Wi-Fi.

Advantages

- Confidentiality.
- Authentication.
- Data integrity.
- Protection against MITM attacks.

Conclusion

Using HTTPS together with a VPN provides secure communication even over public Wi-Fi networks.

- **Question 5: Data Integrity in Cloud Storage**
- **A. Explain how hashing verifies data integrity.**

Hashing converts data into a fixed-length value called a hash.

Working

1. Generate the hash before uploading the file.

2. Store the hash securely.
3. Download the file later.
4. Generate a new hash.
5. Compare both hashes.

If both hashes match, the file has not been modified.